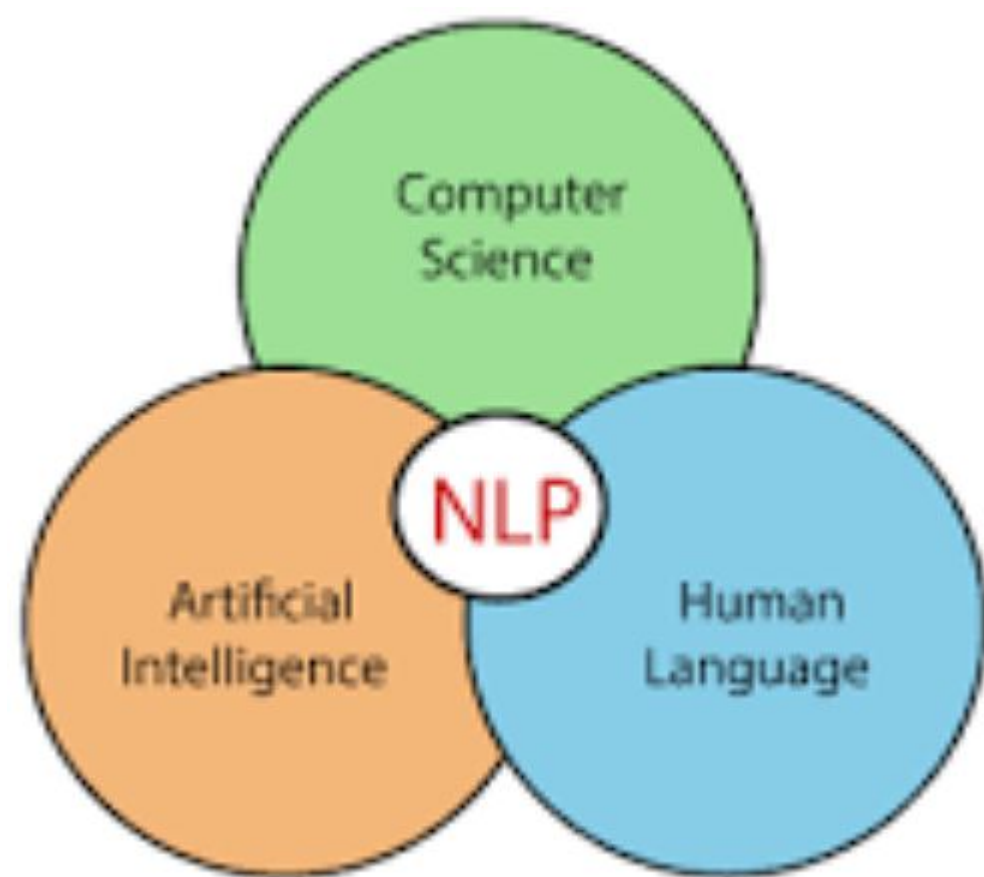


Understanding Natural Language Processing: Basics and Beyond

This presentation delves into the fundamental concepts of Natural Language Processing (NLP), illustrating its key applications, technical challenges, and future trends in the field. Aimed at engineering professionals, it enhances understanding of NLP's critical role within AI and its impact on technological advancements.

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Understanding Natural Language Processing (NLP)

Exploring the fundamentals of NLP and its key components



Definition of NLP

Natural Language Processing is a branch of AI focused on human-computer



Goals of NLP

The primary goal is to enable machines to understand and respond to human language.



Key NLP Tasks

Includes translation, sentiment analysis, and information retrieval among others.



Components of NLP

Key components include syntax, semantics, pragmatics, and discourse.



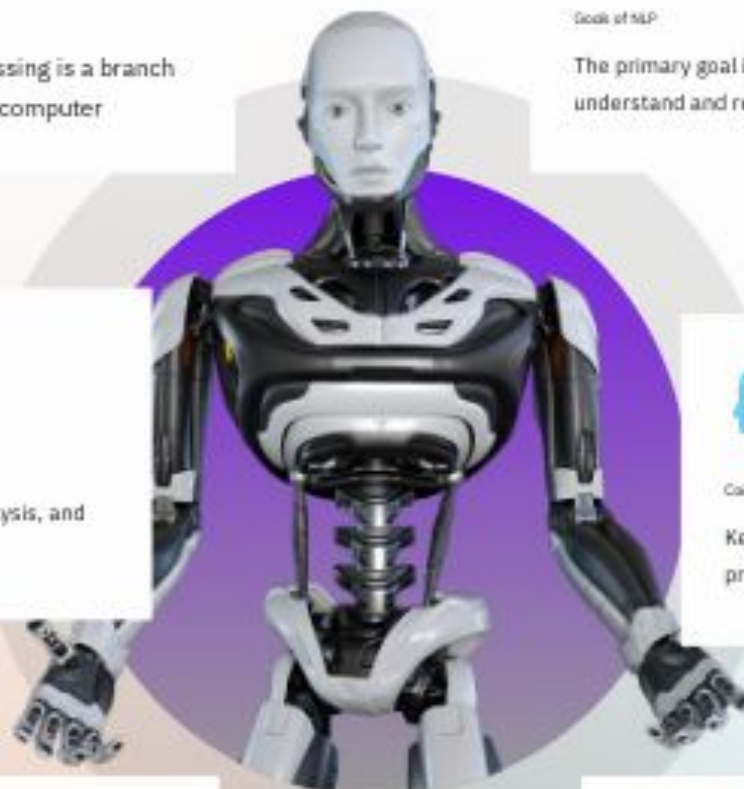
Data Processing in NLP

Involves processing large amounts of language data for meaningful insights.



Human Communication Understanding

NLP aims to enhance how machines comprehend human communication effectively.



What is NLP?

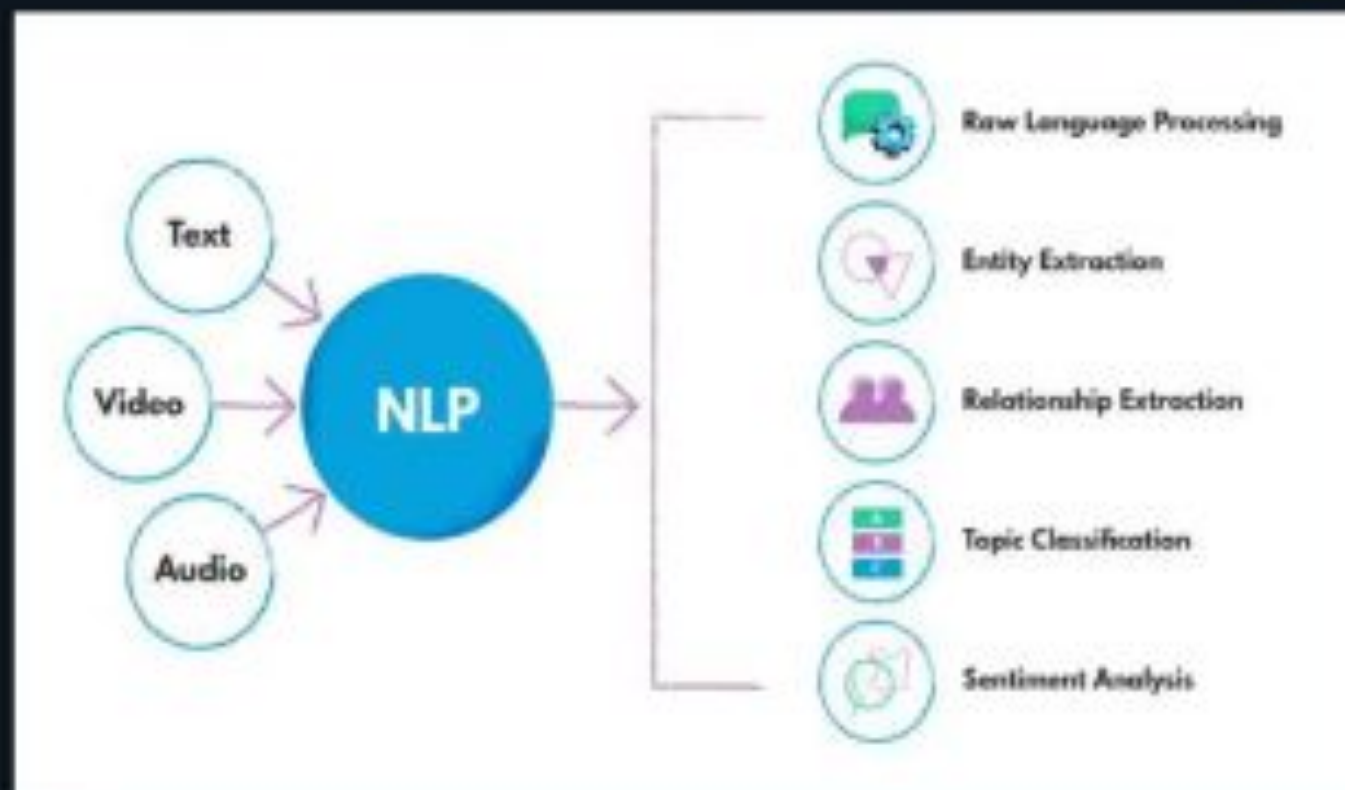
- Ability of a computer program to understand human language as it is spoken and written.
- Gives computers the ability to interpret, manipulate, and comprehend human language.
- Example: Language translation, Search results, Predictive text.



Input and Output of NLP

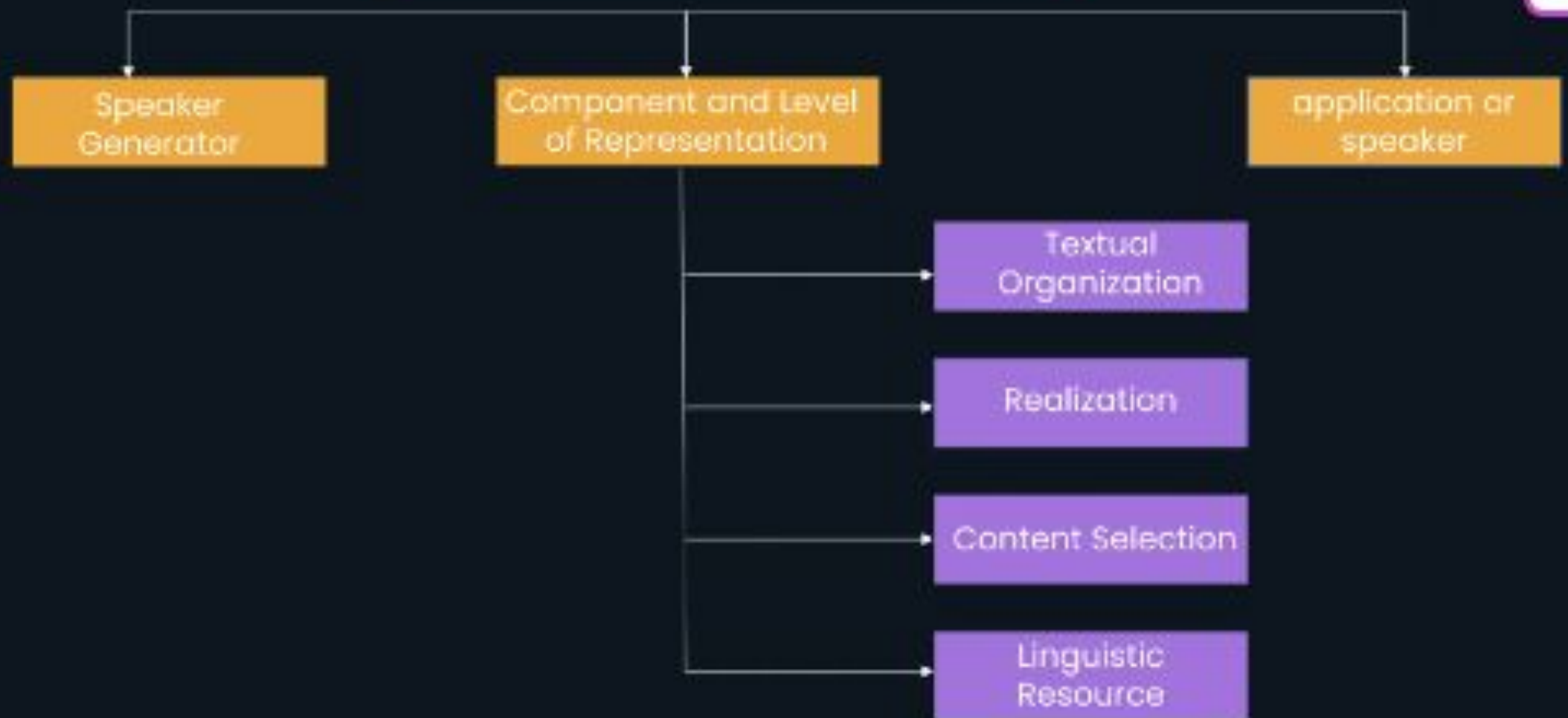
The input and output of an NLP system can be

- Speech
- Written Text



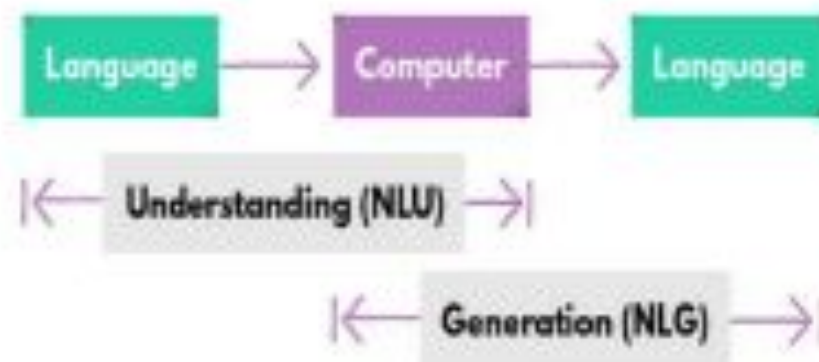
Components of NLP

NATURAL LANGUAGE COMPONENTS



Natural Language Understanding (NLU)

- deals with the ability of computers to **understand human language**
- **Mapping the given input** in natural language into useful representations.
- **Analyzing different aspects** of the language.



Natural Language Generation (NLG)



In many of the most interesting problems in natural language processing, language is the output. Natural language generation focuses on three main scenarios:

It involves –

- **Text planning** – It includes retrieving the relevant content from knowledge base.
- **Sentence planning** – It includes choosing required words, forming meaningful phrases, setting tone of the sentence.
- **Text Realization** – It is mapping sentence plan into sentence structure.

Understanding Technical Challenges in NLP

Exploring key hurdles faced by Natural Language Processing in modern applications

Ambiguity

Ambiguity in language arises when words possess multiple meanings based on their context, making it challenging for NLP systems to interpret correctly and accurately. This issue complicates language processing tasks significantly.



Data Requirements

The effectiveness of NLP models heavily relies on extensive datasets. However, **obtaining large datasets** can be difficult due to privacy concerns, data availability, and resource constraints.



Sarcasm and Irony

Understanding **sarcasm and irony** poses a significant challenge for NLP systems, as these nuanced forms of language require deep contextual understanding and often contradict literal meanings.



Difficulties in NLU

The NLU is harder than NLG.

NL has an extremely rich form and structure.

- **Lexical ambiguity** - It is at very primitive level such as word-level.

For example: treating the word "board" as noun or verb?

- **Syntax Level ambiguity** - A sentence can be parsed in different ways.

For example: "He lifted the beetle with red cap." - Did he use cap to lift the beetle or he lifted a beetle that had red cap?

- **Referential ambiguity** - Referring to something using pronouns. For example, Rima went to Gauri. She said, "I am tired." - Exactly who is tired?

- One input can mean different meanings.
- Many inputs can mean the same thing.

Exploring NLP Applications Across Industries

Understanding the transformative role of NLP in enhancing user experience and efficiency



Chatbots

Enhance customer service with instant responses.



Sentiment Analysis

Gauge customer opinions from social media.



Language Translation

Facilitate communication across languages.



Healthcare

Extract insights from patient records.

Why **NLP** matters?

Communication

Enables real-time language translation and chatbots.

01



Sentiment Analysis

Understands opinions and emotions in social media and customer reviews.

04

Information Retrieval

Enhances search engines and information extraction.

02

Content Creation

Assists in generating human-like text for various purposes.

03

Benefits of natural language processing

- By enabling computers to understand human language, interacting with computers becomes much more intuitive for humans.
- improved accuracy and efficiency of documentation;
- ability to automatically make a readable summary of a larger, more complex original text;
- useful for personal assistants such as Alexa, by enabling it to understand spoken word;
- enables an organization to use chatbots for customer support;
- easier to perform sentiment analysis; and
- provides advanced insights from analytics that were previously unreachable due to data volume.

Steps of NLP

Lexical Analysis -

- Identifying and analyzing the structure of words.
- dividing the whole chunk of txt into paragraphs, sentences, and words.

Syntactic Analysis (Parsing) -

- analysis of words in the sentence for grammar
- arranging words in a manner that shows the relationship among the words.
- The sentence such as "The school goes to boy" is rejected by English syntactic analyzer.
- CFG and Parsing.

Semantic Analysis -

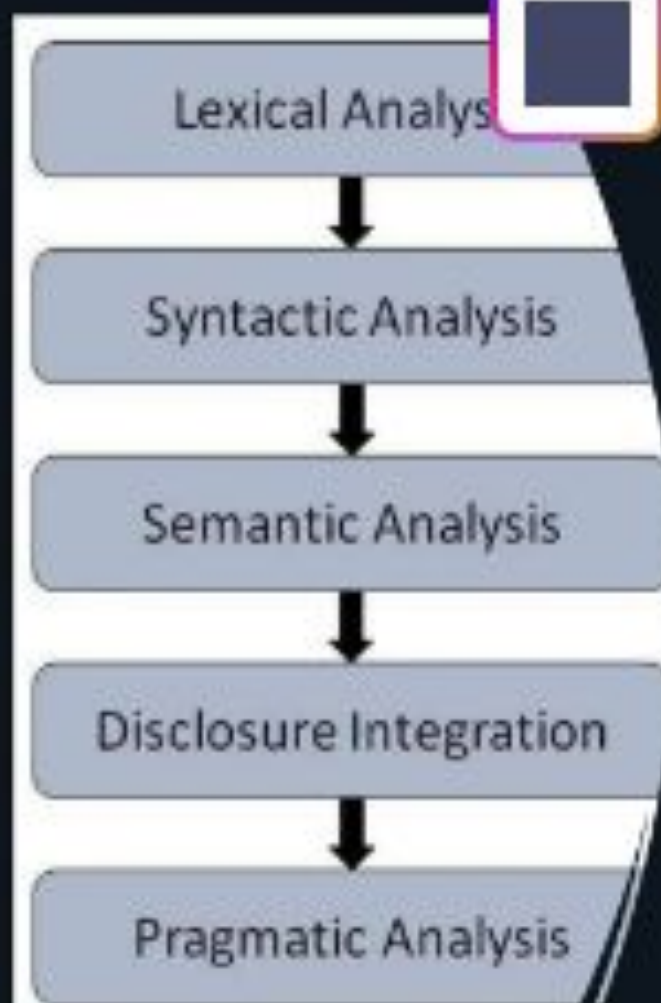
- The text is checked for meaningfulness.
- The semantic analyzer disregards sentence such as "hot ice-cream".

Discourse Integration -

- The meaning of any sentence depends upon the meaning of the sentence just before it. In addition, it also brings about the meaning of immediately succeeding sentence.

Pragmatic Analysis -

- what was said is re-interpreted on what it actually meant.



NLP Pipeline



Text Preprocessing

Tokenization, Lowercasing,
Removing Noise,
Stopword Removal

Feature Extraction

N-grams
POS Tagging
Named Entity Recognition (NER)

Machine Learning Model

Classification, Sentiment Analysis,
Sequence Labeling, Language Translation
Text Generation



Text Representation

Word Embeddings
TF-IDF
Document Embeddings

Text Understanding

Syntax and Parsing
Semantic Analysis
Coreference Resolution

Post-processing

Lemmatization/Stemming
Spell Checking
Entity Resolution
Result Interpretation

What is Morphology?

- Study of Words
 - Their **internal structure**

washing → wash + -ing

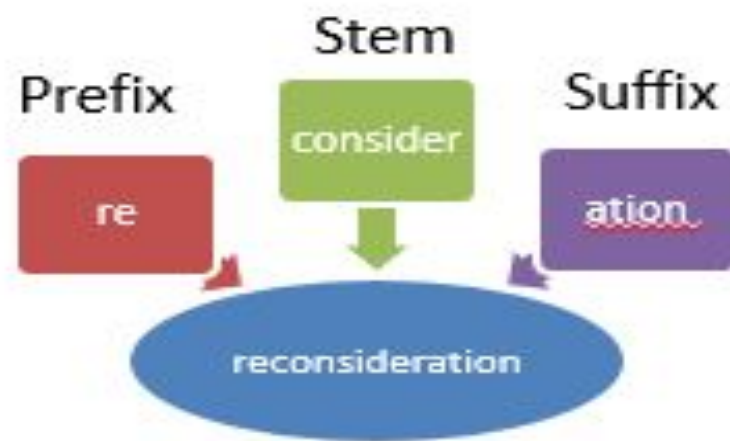
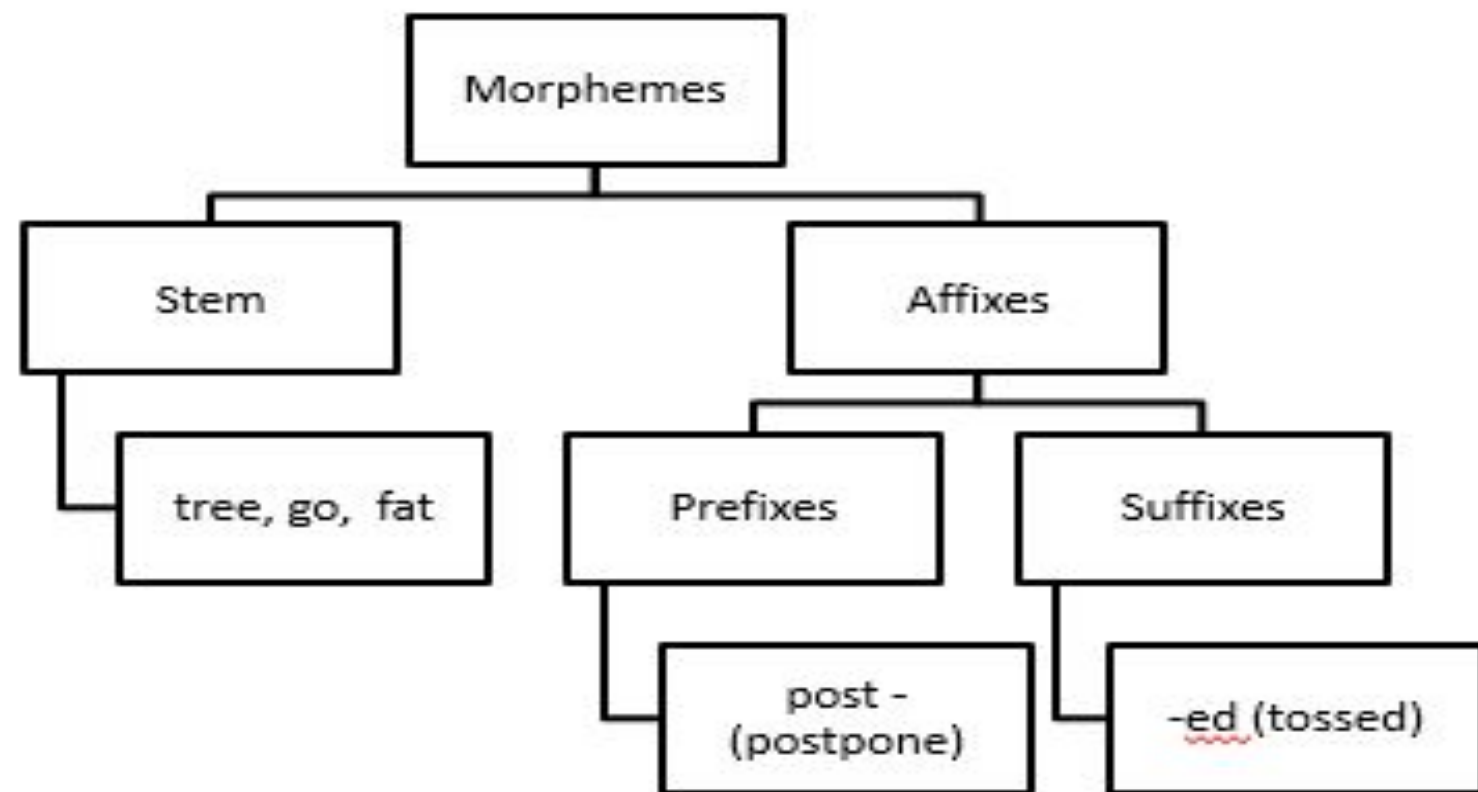
- How they are **formed**?

bat	→	bats	∴	rat	→	rats
write	→	writer	∴	browse	→	browser

- Morphology tries to formulate rules

Morphemes

- Smallest meaning bearing units constituting a word



Classes of Morphology

Inflection

Indicates some grammatical function.

Results in a word of the same class

Derivation

- Usually, results in a word of a different class
 - -able when attached to a verb gives an adjective
 - read (V) + -able = readable (Adj)
- Often meaning of the derived word is difficult to predict exactly
 - writer :: writer (one who writes)
 - paint :: painter (one who paints)
 - cut :: cutter? (an instrument used to cut)
 - Less productive
 - eatable :: readable :: runnable?

The Revolutionary Impact of Machine Learning on NLP

Exploring how machine learning has transformed natural language processing through advanced techniques and models.



Machine learning enhances NLP accuracy significantly.

The integration of **machine learning** into **NLP** has led to remarkable improvements in the accuracy of various tasks, enabling more effective communication between humans and machines.



Deep learning models improve contextual understanding.

Advanced algorithms like **deep learning** allow for superior **feature extraction** and **contextual understanding**, making applications in NLP more sophisticated and human-like in their responses.



Neural networks enable advanced language applications.

Techniques such as **neural networks** have paved the way for complex applications in **NLP**, including text classification, translation, and sentiment analysis, by leveraging vast datasets.



Innovative models like BERT and GPT-3.

Models like **BERT** and **GPT-3** utilize extensive datasets to learn about language structure and nuances, allowing them to generate human-like text and responses in various applications.

Natural Language Toolkit (NLTK)

- A collection of Python programs, modules, data set and tutorial to support research and development in Natural Language Processing (NLP)
- Written by Steven Bird, Edward Loper and Ewan Klein
- NLTK is
 - *Free and Open source*
 - *Easy to use*
 - *Modular*
 - *Well documented*
 - *Simple and extensible*

Installing NLTK Data

- NLTK comes with many corpora, toy grammars, trained models, etc. A complete list is posted at: http://nltk.org/nltk_data/
- Run the Python REPL and type the commands:

```
>>> import nltk  
>>> nltk.download()
```

- A new window should open, showing the NLTK Downloader. Click on the File menu and select Change Download Directory. For central installation, set this to C:\nltk_data (Windows), /usr/local/share/nltk_data (Mac), or /usr/share/nltk_data (Unix). Next, select the packages or collections you want to download.

- Tokenization
- Corpora
- Frequency Distribution
- Stylistics
- Sentence Tokenization
- WordNet
- Stemming
- Lemmatization
- Part of Speech Tagging
- Tagging Methods

- Unigram Tagging
- N-gram Tagging
- Chunking – Shallow Parsing
- Entity Recognition
- Supervised Classification
- Document Classification
- Hidden Markov Models - HMM
- References

Shallow VS deep parsing

- In **deep** or **full parsing**, typically, grammar concepts such as **CFG**, and probabilistic context-free grammar (**PCFG**), and a search strategy is used to give a complete syntactic structure to a sentence.
- **Shallow parsing** is the task of parsing a **limited** part of the syntactic information from the given text.

The two approaches in parsing

The rule-based approach	The probabilistic approach
This approach is based on rules/grammar	In this approach, you learn rules/grammar by using probabilistic models
Manual grammatical rules are coded down in CFG, and so on, in this approach	This uses observed probabilities of linguistic features
This has a top-down approach	This has a bottom-up approach
This approach includes CFG and Regexbased parser	This approach includes PCFG and the Stanford parser